Mode stability of subextremal Kerr–de Sitter black holes

Teukolsky ($s = \pm 2$) and ungauged linearised Einstein equations: the proof as it stands

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Abstract

For every subextremal Kerr–de Sitter (KdS) black hole we prove (A) that the spin ± 2 Teukolsky equation has no nonzero future-regular mode solutions with $\operatorname{Im}\sigma \geq 0$, apart from the allowed stationary ones, and (B) that every generalised mode solution of the ungauged linearised Einstein equation $D\mathrm{Ric}(h) = \Lambda h$ with $\operatorname{Im}\sigma \geq 0$ is pure gauge if $\sigma \neq 0$, and a linearised KdS metric plus pure gauge if $\sigma = 0$. Statement (B) is the hypothesis of the full-subextremal nonlinear stability theorem announced by Hintz–Petersen–Vasy. **Status.** Part (B) is proved, given (A). Part (A) is proved except for one explicit inequality, condition (D) of the event lemma in the large-A regime R3 (Section 2.3), which is verified numerically with margin but is not yet proved. Every step marked *[cert]* uses an exact symbolic or a short numerical verification; the scripts are named in the repository ledger `kds/LEMMAS.md` (item numbers below).

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1 Setting and statements

KdS with parameters (Λ, M, a), $\Lambda > 0$, in Boyer–Lindquist coordinates has $\Delta_r = (r^2 + a^2)(1 - \Lambda r^2/3) - 2Mr = -\frac{\Lambda}{3}\prod_{j=0}^{3}(r - r_j)$, $\alpha = \Lambda a^2/3$, $\Xi = 1 + \alpha$. *Subextremal* means four distinct real roots $r_3 < 0 < r_0 < r_1 = r_e < r_2 = r_c$; then $\sum r_j = 0$. Modes are $e^{-i\sigma t + im\varphi}$ times functions of (r, θ), smooth across both horizons in the regular time t_*.

Theorem 1 (A). *For every subextremal KdS the spin $s = \pm 2$ Teukolsky equation has no nonzero future-regular mode with $\operatorname{Im}\sigma \geq 0$, except those with $\sigma = 0$ permitted below.*

Theorem 2 (B). *Let h be a generalised mode, $(L_T + i\sigma)^k h = 0$ with $\operatorname{Im}\sigma \geq 0$, of $D\mathrm{Ric}(h) = \Lambda h$, smooth across both horizons. If $\sigma \neq 0$ then $h = \mathcal{L}_X g$ with X smooth; if $\sigma = 0$ then $h = c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$ with X polynomial in t_*.*

2 Part A: Teukolsky mode stability

2.1 Heun form and the window

The Möbius map $r_0, r_1, r_2, r_3 \mapsto 0, 1, t, \infty$ puts the radial equation in Heun form with $\mu(z) = z(z-1)(t-z)$, $t = \frac{(r_1-r_3)(r_2-r_0)}{(r_1-r_0)(r_2-r_3)} > 1$ (item 26). With $K(r) = \omega(r^2+a^2) - am$ and $\kappa_j = i\Xi K(r_j)/\Delta_r'(r_j)$, put $e_1 = 2(\kappa_1+\kappa_3)$, $e_t = 2(\kappa_2+\kappa_3)$. For real ω the dual equation is real and the Wronskian identity gives $\operatorname{Im} e_1\, |c_1|^2 = t \operatorname{Im} e_t\, |c_t|^2$ (item 34).

Proposition 3 (the window; item 88). *Let $m \neq 0$, $w = \omega/m$, $\rho_j = r_j^2 + a^2$, $\Omega_j = a/\rho_j$, $D_j = |\Delta_r'(r_j)|$, $P_j = \rho_j/D_j$. Then $\operatorname{Im} e_1 = 2\Xi m f_1(w)$, $\operatorname{Im} e_t = 2\Xi m f_2(w)$ with $f_1 = P_1(w-\Omega_1) + P_3(w-\Omega_3)$ increasing and $f_2 = -P_2(w-\Omega_2) + P_3(w-\Omega_3)$ decreasing; their roots satisfy $0 < w_2 < w_1$. Hence $\operatorname{Im} e_1 \cdot \operatorname{Im} e_t > 0$ exactly on the open window (w_2, w_1).*

Proof. $\operatorname{sign}\Delta_r'(r_j) = (-,+,-,+)$. Since ρ/Δ_r and $1/\Delta_r$ have zero residue at infinity, $P_1 + P_3 = P_0 + P_2$ and $D_1^{-1} + D_3^{-1} = D_0^{-1} + D_2^{-1}$. So the slope of f_2 is $P_0 - P_1$, and $P_1 > P_0$ because $D_0/D_1 = f(r_0)/f(r_1) > 1$ for the concave parabola $f(x) = (x-r_3)(r_2-x)$ with vertex $-(r_0+r_1)/2 < r_0$, while $\rho_1 > \rho_0$. This also gives $w_2 = a(D_1^{-1} - D_0^{-1})/(P_1 - P_0) > 0$. Finally, cross-multiplying $w_2 < w_1$, the a^2-terms cancel and $D_0 D_1 D_3$ times the difference factors as $2(r_0-r_1)(r_0-r_3)(r_1-r_3)(r_0 r_3 - r_1 r_2) > 0$. □

Corollary 4. *(i) Outside the window, and for $m = 0$, $\omega \neq 0$ real, there is no real-frequency mode. (ii) $\omega = 0$ lies outside the window for $m \neq 0$. (iii) At w_2 only $\operatorname{Im} e_t$ vanishes and at w_1 only $\operatorname{Im} e_1$; both endpoints are then closed by the Wronskian and the injectivity of the transform (item 44).*

2.2 Inside the window: transform, dual potential, event lemma

Inside the window the Riemann–Liouville transform $I_1^{\mu+1}$ maps a quasinormal mode injectively to a solution of a dual Heun equation with the correct boundary branches (items 17, 55). The dual potential is exactly

$$L(z) = \frac{A}{z} + B - 3z, \qquad A = t\big(X_0^2 - (X_1 + X_t)^2\big) > 0, \qquad B = \lambda C_0 + c_1,$$

with $C_0 = 3/(\Lambda(r_1 - r_0)(r_2 - r_3))$ and $c_1 = 1 + t + 3(1-\alpha)C_0$.

Lemma 5 (the c_1 identity; item 89, *cert*). *With $\Lambda = 3$ (by scaling) and the exact relations $\sum \kappa_j = 0$, $\sum r_j^2 \kappa_j = 0$, $\sum r_j \kappa_j = -i\Xi\omega$ (residue identities), as rational functions $n_1 - (\lambda C_0 + c_1) = -2(s^2 - 4)\frac{r_0^2 + r_0 r_1 + r_0 r_2 - r_1 r_2}{(r_0 - r_1)(r_0 + r_1 + 2r_2)}$ and $n_2 + 3 = 4 - s^2$.*

The angular bound $\lambda \geq 3/2$ on the window (item 46) and $C_0 > (2t-1)/(3(1-\alpha))$ (item 72, exact form $C_0 - \frac{3}{2} \cdot \frac{2t-1}{3(1-\alpha)} = \frac{3(r_0+r_1)^2}{2\Lambda(r_1-r_0)(r_2-r_3)}$) give the floor $B \geq 5.5t - 1.25$.

The two-current argument (items 8, 12, 17) reduces mode stability inside the window to the following.

Definition 6 (event lemma)**.** For $t > 1$, $z_0 \in (1, t)$, $A > 0$, $B \geq 5.5t - 1.25$, find h on $(1, z_0)$ with $s = z_0 - z$, $H = 2\mu + s\mu'$ and

$$\text{(C)} \quad s\,h\,\mu \leq H, \qquad \text{(D)} \quad \tfrac{1}{2}\big(\mu' h + \mu(h' + h^2)\big) \leq s(Lh + L').$$

Regimes R1 ($p \leq 1$) and R2 ($p > 1$, $2(z_0-1)^2(A/z_0+3) < t-1$), with $p = 2(z_0-1)L(1)/(t-1)$, are closed by the explicit ODE multiplier of item 47. In R3 we use the rational multiplier (item 104)

$$h = 1.7\,g\,\frac{w}{1+w} + \frac{0.45\,(1-u)}{xz}, \quad g = -\frac{L'}{L},\ w = 8\kappa s^2,\ \kappa = \frac{L(z_0)}{\mu(z_0)},\ x = z - 1,\ u = \frac{x}{z_0 - 1}.$$

Proposition 7 ((C) in R3; item 105)**.** *(C) holds for all t, z_0, A and $B \geq 5.5t - 1.25$.*

Proof. All three factors of h are nonnegative, $w/(1+w) < 1$, $1 - u \leq 1$, so it suffices that $1.7sg + 0.45s/(xz) \leq 2 + s/z + s/x - s/(t-z)$ (divide by μ). Now $g - 1/z = (6 - B/z)/L$, so $g \leq 1/z$ when $B \geq 6z$ and otherwise the excess is at most $(0.5 + 1.25/t)/(2.5t - 1.25)$; $0.7s/z + 0.45s/(xz) - s/x \leq -0.3s/z$; $s/(t-z) < 1$; and where the excess is nonzero $1.7s \cdot \text{excess} \leq 0.69$. Summing, the difference is at most $-0.3s/z + 1 + 0.69 - 2 < 0$. □

2.3 The remaining open condition

Open (D) in R3. Condition (D) for the rational multiplier on R3.

Evidence and reductions (items 104–110): on a 1200-case grid and on 3×10^5 random R3 points the normalised margin is ≤ -0.028. Splitting by w: in the bulk $w \geq 10$, (D) follows from the cutoff-free inequality

$$\text{(B3)} \quad \tfrac{1}{2}\mu' h_{\pm} + 0.85\mu g^2 + \tfrac{1}{2}\mu(1.7g + q)^2 + \tfrac{1}{2}\mu q' \leq 0.5455\, sLg + sLq,$$

which holds numerically with margin -0.11. For fixed (t, z_0, A, z) the relevant expressions are decreasing in B and all constraints are monotone in B, so B may be put at the floor (item 108); then the identity $g = 1/z - (B - 6z)/(zL)$ makes (B3) a function of the single variable $L \in [L_{\min}, \infty)$ (item 109). In the seam zone $w \leq 10$ the cutoff derivative is load-bearing and the problem rescales to the layer problem of items 13–15. What is missing is a proof (by hand or by certificate) of (B3) in these reduced variables and of the seam zone.

2.4 $\sigma = 0$ and the upper half plane

For $m = 0$, $\sigma = 0$: an energy identity with $V = -2\Lambda r^2 - 2(1-\alpha) - \lambda < 0$ for $s = -2$, and for $s = +2$ the Teukolsky–Starobinsky map, whose kernel is excluded by $\det = [\lambda^2 + 6(1-\alpha)\lambda + 8 - 4\alpha + 8\alpha^2]^2 > 0$ (items 35, 43). Modes with $\operatorname{Im}\sigma > 0$ are excluded by continuity from Schwarzschild–de Sitter: QNMs depend continuously on (Λ, a) (Petersen–Vasy, full subextremal Fredholm theory) and none can cross the real axis by the above (item 36).

3 Part B: ungauged linearised Einstein mode stability

Throughout, $a \neq 0$; for $a = 0$ Theorem B is Hintz–Vasy 2018. By Theorem A, $\delta\Psi_0 = \delta\Psi_4 = 0$ for every generalised mode (item B.1). We use GHP with $\text{Þ} = D - p\epsilon - q\bar{\epsilon}$, $\eth = \delta - p\beta - q\bar{\alpha}$, $\text{Þ}' = \Delta - p\gamma - q\bar{\gamma}$, $\eth' = \bar{\delta} - p\alpha - q\bar{\beta}$, $\rho' = -\mu$, $\tau' = -\pi$ (conventions fixed by machine checks on KdS, item 92). On KdS, $\Psi_2 = -M/(r - ia\cos\theta)^3$; put $\phi = r - ia\cos\theta \propto \Psi_2^{-1/3}$.

3.1 Linearised Goldberg–Sachs

Lemma 8 (item 90). *Infinitesimal null rotations achieve $\delta\Psi_1 = \delta\Psi_3 = 0$; then $\delta\kappa = \delta\sigma = \delta\kappa' = \delta\sigma' = 0$.*

Proof. To first order a null rotation about n with parameter d shifts $\delta\Psi_1$ by $3d\Psi_2$ and fixes $\delta\Psi_0, \delta\Psi_2$; about l similarly for $\delta\Psi_3$. As $\Psi_2 \neq 0$ the choice is algebraic. The vacuum-Λ Bianchi identities have no Λ terms; linearising $\text{Þ}\psi_1 - \eth'\psi_0 = -\tau'\psi_0 + 4\rho\psi_1 - 3\kappa\psi_2$ gives $\delta\kappa = 0$, and $\text{Þ}'\psi_0 - \eth\psi_1 = \rho'\psi_0 - 4\tau\psi_1 + 3\sigma\psi_2$ gives $\delta\sigma = 0$; primes give the rest. □

3.2 The Killing-spinor field and reality

On a type D Λ-vacuum (with $\kappa = \sigma = \kappa' = \sigma' = 0$) the GHP derivatives of the tetrad are as listed in item 93(b) and the following hold (item 91; machine-checked on KdS and on the accelerating NUT Plebański–Demiański metric, items 92, 97): $\text{Þ}\rho = \rho^2$, $\eth\rho = (\rho - \bar\rho)\tau$, $\text{Þ}\tau = (\tau - \bar\tau')\rho$, $\eth\tau = \tau^2$ and primes, $\text{Þ}\rho' - \text{Þ}'\rho = \tau\bar\tau - \tau'\bar\tau'$, $\eth\tau' - \eth'\tau = \rho\bar\rho' - \bar\rho\rho'$, $\text{Þ}\tau' = \eth'\rho$, $\text{Þ}'\tau = \eth\rho'$.

Lemma 9 (items 93, 96). *With $A = \phi\rho$, $B = -\phi\rho'$, $C = \phi\tau$, $D = -\phi\tau'$, the field $\xi = Bl + An - Dm - C\bar m$ satisfies all ten Killing equations as an algebraic consequence of these identities. On KdS, $\xi = -\Xi\,\partial_t$.*

Hence for a perturbation in the gauge above, $\mathcal{L}_\xi h + \mathcal{L}_{\delta\xi} g = 0$ with $\delta\xi$ a local linear expression in h commuting with $\mathcal{L}_{\partial_t}, \mathcal{L}_{\partial_\varphi}$.

Lemma 10 (reality for modes; item 96). *Let $J[h] = (\delta\xi[h] - \overline{\delta\xi[\bar h]})/(2i)$. Then $\mathcal{L}_{J[h]} g = 0$, so $J[h]$ is a constant combination of $\partial_t, \partial_\varphi$ (the local Killing algebra of KdS with $a \neq 0$ is two-dimensional, item 97 R2). For a mode with $(\sigma, m) \neq (0,0)$, $J[h]$ is a mode of the same type and so vanishes.*

Consequently, to first order, $\phi\rho = \bar\phi\bar\rho$, $\phi\rho' = \bar\phi\bar\rho'$, $\phi\tau = -\bar\phi\bar\tau'$.

3.3 The second Killing field

Put $R = \operatorname{Re}\phi$, $I = \operatorname{Im}\phi$ (the Carter coordinates r, $a\cos\theta$). Reality gives $\text{Þ}R = -A$, $\text{Þ}'R = B$, $\eth R = \eth' R = 0$, $\text{Þ}I = \text{Þ}'I = 0$, $\eth I = iC$, $\eth' I = -iD$.

Lemma 11 (items 96, 97). *$\zeta = I^2(Bl + An) + R^2(Dm + C\bar m)$ satisfies all ten Killing equations by algebra (the ln, $m\bar m$, lm, nm components are computed in item 96–97 and cancel by reality; the others are conjugates). On KdS $\zeta = -\Xi(a^2\partial_t + a\partial_\varphi)$.*

3.4 Conclusion for $(\sigma, m) \neq (0,0)$

From $\mathcal{L}_\xi h = -\mathcal{L}_{\delta\xi} g$ and $\mathcal{L}_\zeta h = -\mathcal{L}_{\delta\zeta} g$: if $\sigma \neq 0$, $h = \mathcal{L}_X g$ with $X = \delta\xi/(i\sigma\Xi)$; if $\sigma = 0$, $m \neq 0$, $h = \mathcal{L}_X g$ with $X = -\delta\zeta/(im\Xi a)$. X is smooth across the horizons because $\xi, \zeta, \delta\xi, \delta\zeta$ are boost-spin invariant natural constructions from the Weyl spinor (item 97 R3). HPV's modes with zero T-frequency and $m \neq 0$ have ∂_t-frequency $\Omega m \neq 0$.

3.5 $\sigma = m = 0$

Proposition 12 (items 98, 102, 103). *A stationary axisymmetric h with $\delta\Psi_0 = \delta\Psi_4 = 0$ is $c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$.*

Sketch. The constants in $J[h]$ are removed by a phase of ϕ and by the acceleration tangent, which is the only Plebański–Demiański tangent making ξ complex (item 95, exact nonlinear check). Then ξ, ζ are first-order Killing fields, they commute, $\nabla R \perp \nabla I$, $\Delta_r := -2AB|\phi|^2$ depends on R only and $\Delta_x := -2CD|\phi|^2$ on I only, and in adapted coordinates the perturbed metric is Carter's ansatz to first order. On that ansatz the Λ-vacuum equations are linear in (Δ_r, Δ_x), with solutions exactly $\Delta_r = -\frac{\Lambda}{3}R^4 + \varepsilon R^2 - 2mR + \gamma$, $\Delta_x = -\frac{\Lambda}{3}I^4 - \varepsilon I^2 + 2nI + \gamma$ (item 103, *cert*). Modulo the scaling, the variations are $\delta M, \delta a, \delta n$; with acceleration, four. Axis regularity (determinant $-64mp_0^4/c \neq 0$) removes NUT and acceleration. □

3.6 Generalised modes

For $(\sigma, m) \neq (0, 0)$, induction on k: the top coefficient is a mode, hence $\mathcal{L}_{X_1} g$, and $Y = t_*^{k-1} X_1/(k-1)!$ lowers the order. For $\sigma = m = 0$, linear growth of $(\delta M, \delta a)$ is excluded by the Iyer–Wald charges: they are surface-independent, vanish on every $\mathcal{L}_X g$ by covariance of the Noether charge (item 101), and $\partial(E, J)/\partial(M, a)$ has determinant $M/\Xi^4 \neq 0$ for $E = M/\Xi^2$, $J = aM/\Xi^2$ (item 100).

4 What remains

1. **(D) in R3** (Section 2.3): the only open mathematical item in Theorem A, hence in both theorems.

2. An independent check of Part B (the in-house hostile passes are items 97 and 102).